KAIZEN MATHS

Lesson starter

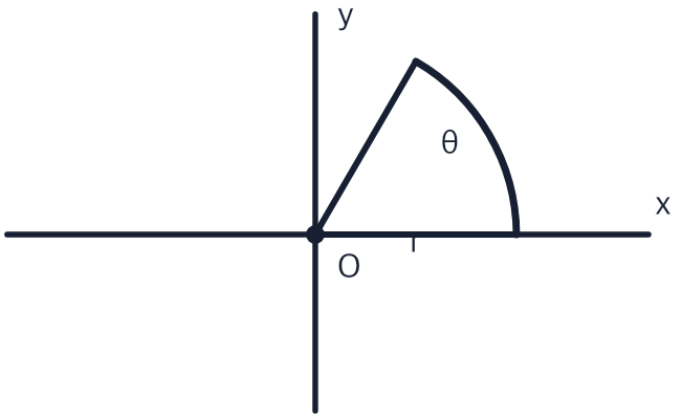
Polar Coordinates and Polar Curves

Area in Polar Coordinates

Name _____

SA1

Find the area of each polar sector.



Find the area of the sector bounded by $r = 2$, $\theta = 0$, and $\theta = \pi/3$.

Show your working

KAIZEN MATHS

Lesson starter

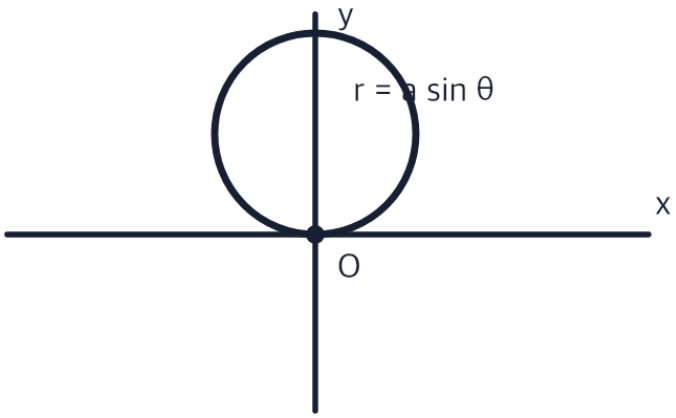
Polar Coordinates and Polar Curves

Area in Polar Coordinates

Name _____

SB1

Find the area enclosed by each polar circle.



Find the area enclosed by $r = 6 \sin \theta$.

Show your working

KAIZEN MATHS

Lesson starter

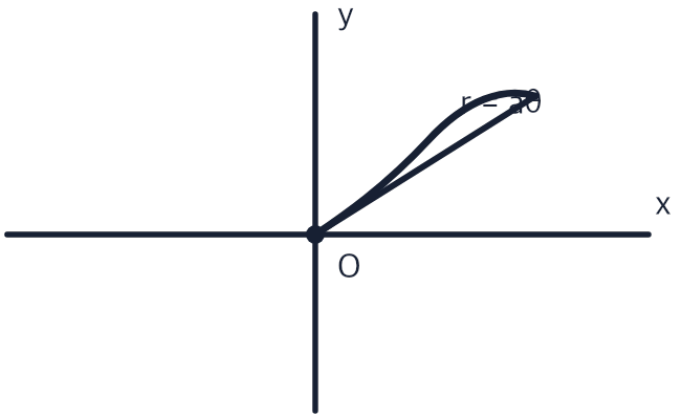
Polar Coordinates and Polar Curves

Area in Polar Coordinates

Name _____

SC1

Use the polar area formula to find the area.



Find the area bounded by $r = 3\theta$, $\theta = 0$, and $\theta = \pi/2$.

Show your working

KAIZEN MATHS

Lesson starter

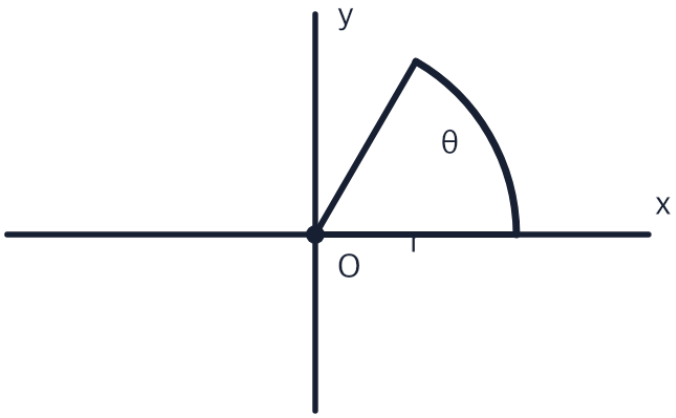
Polar Coordinates and Polar Curves

Area in Polar Coordinates

Name _____

SD1

Find the area of each polar sector.



Find the area of the sector bounded by $r = 5$, $\theta = 0$, and $\theta = \pi/6$.

Show your working

KAIZEN MATHS

Exit ticket

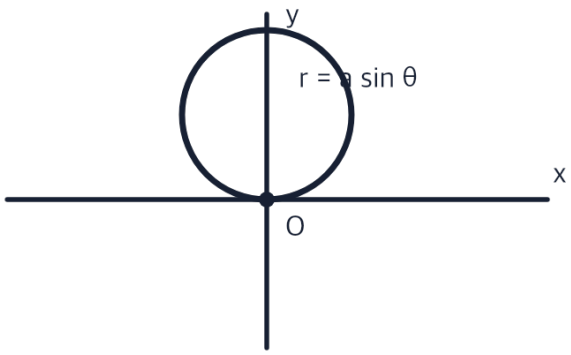
Polar Coordinates and Polar Curves

Area in Polar Coordinates

Name _____

EA1

Find the area enclosed by each polar circle.



Find the area enclosed by r = 8 sin θ.

Show your working

KAIZEN MATHS

Exit ticket

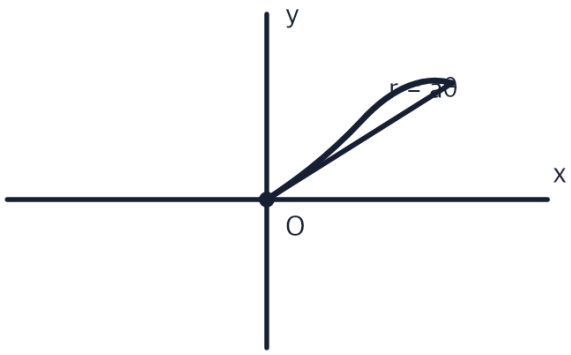
Polar Coordinates and Polar Curves

Area in Polar Coordinates

Name _____

EB1

Use the polar area formula to find the area.



Find the area bounded by r = 2θ, θ = 0, and θ = π.

Show your working

KAIZEN MATHS

Exit ticket

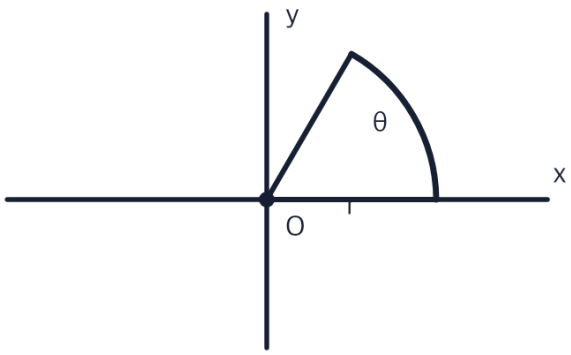
Polar Coordinates and Polar Curves

Area in Polar Coordinates

Name _____

EC1

Find the area of each polar sector.



Find the area of the sector bounded by r = 6, θ = 0, and θ = π/6.

Show your working

KAIZEN MATHS

Exit ticket

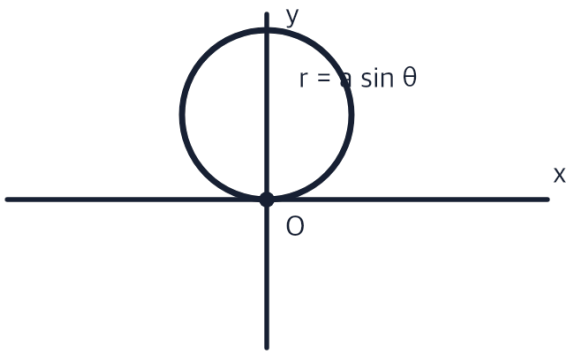
Polar Coordinates and Polar Curves

Area in Polar Coordinates

Name _____

EA1

Find the area enclosed by each polar circle.



Find the area enclosed by r = 8 sin θ.

Show your working

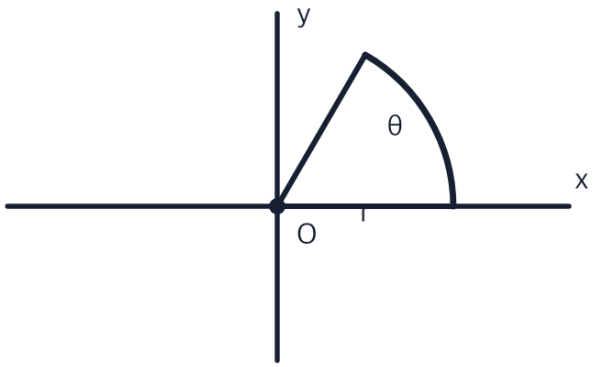
Teacher answers & working

Polar Coordinates and Polar Curves · Area in Polar Coordinates

ALG-294F-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1

Find the area of each polar sector.



Find the area of the sector bounded by r = 2, θ = 0, and θ = π/3.

Answer:

$\frac{2}{3}\pi$

Working:

Use $A = \frac{1}{2}\int r^2\, d\theta$.

Here r = 2, so $r^2 = 4$.

$A = \frac{1}{2}\int_0^{\pi/3} 4\, d\theta$.

$A = \frac{1}{2} \times 4 \times \pi/3$.

Therefore $A = \frac{2}{3}\pi$.

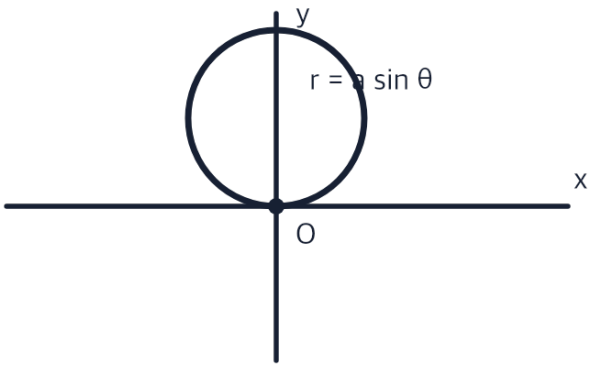
Teacher answers & working

Polar Coordinates and Polar Curves · Area in Polar Coordinates

ALG-294F-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1

Find the area enclosed by each polar circle.



Find the area enclosed by r = 6 sin θ.

Answer:

9π

Working:

Use $A = \frac{1}{2}\int r^2\, d\theta$.

For r = 6 sin θ, use limits 0 to π.

$A = \frac{1}{2}\int (6\sin\theta)^2\, d\theta$.

$A = \frac{36}{2}\int \sin^2\theta\, d\theta$ over the full circle.

The integral of $\sin^2\theta$ from 0 to π is π/2.

So $A = \frac{36}{2} \times \frac{\pi}{2} = 9\pi$.

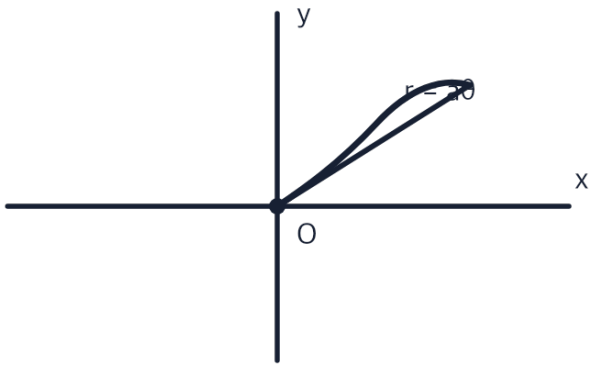
Teacher answers & working

Polar Coordinates and Polar Curves · Area in Polar Coordinates

ALG-294F-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1

Use the polar area formula to find the area.



Find the area bounded by r = 3θ, θ = 0, and θ = π/2.

Answer:

$\frac{3}{16}\pi^3$

Working:

Use $A = \frac{1}{2}\int r^2\,d\theta$.

Here r = 3θ, so $r^2 = 9\theta^2$.

$A = \frac{1}{2}\int_0^{\pi/2} 9\theta^2\,d\theta$.

$A = \frac{9}{2} \times \left[\frac{\theta^3}{3}\right]_0^{\pi/2}$.

$A = \frac{9}{6} \times (\pi/2)^3$.

Therefore $A = \frac{3}{16}\pi^3$.

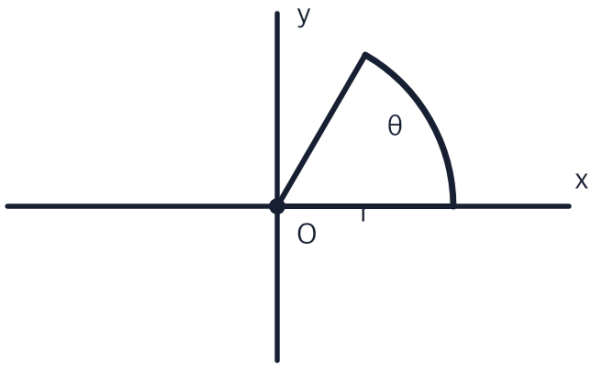
Teacher answers & working

Polar Coordinates and Polar Curves · Area in Polar Coordinates

ALG-294F-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1

Find the area of each polar sector.



Find the area of the sector bounded by r = 5, θ = 0, and θ = π/6.

Answer:

$\frac{25}{12}\pi$

Working:

Use $A = \frac{1}{2}\int r^2\, d\theta$.

Here r = 5, so $r^2 = 25$.

$A = \frac{1}{2}\int_0^{\pi/6} 25\, d\theta$.

$A = \frac{1}{2} \times 25 \times \pi/6$.

Therefore $A = \frac{25}{12}\pi$.

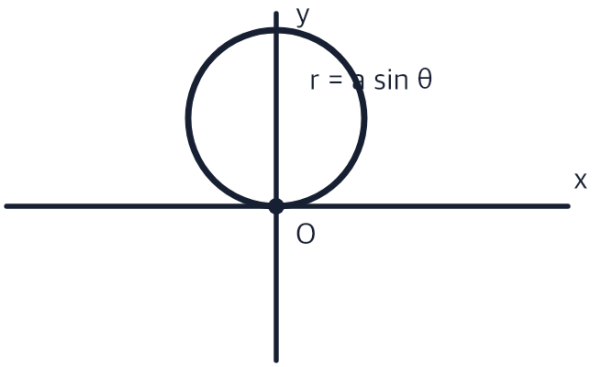
Teacher answers & working

Polar Coordinates and Polar Curves · Area in Polar Coordinates

ALG-294F-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1

Find the area enclosed by each polar circle.



Find the area enclosed by r = 8 sin θ.

Answer:

16π

Working:

Use $A = \frac{1}{2}\int r^2\, d\theta$.

For r = 8 sin θ, use limits 0 to π.

$A = \frac{1}{2}\int (8 \sin \theta)^2\, d\theta$.

$A = \frac{64}{2}\int \sin^2 \theta\, d\theta$ over the full circle.

The integral of $\sin^2 \theta$ from 0 to π is π/2.

So $A = \frac{64}{2} \times \frac{\pi}{2} = 16\pi$.

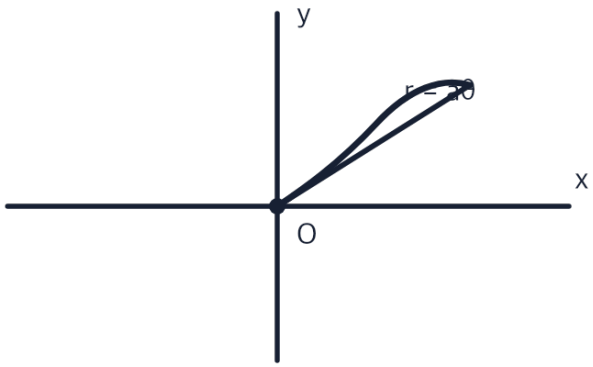
Teacher answers & working

Polar Coordinates and Polar Curves · Area in Polar Coordinates

ALG-294F-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1

Use the polar area formula to find the area.



Find the area bounded by r = 2θ, θ = 0, and θ = π.

Answer:

$\frac{2}{3}\pi^3$

Working:

Use $A = \frac{1}{2}\int r^2\,d\theta$.

Here r = 2θ, so $r^2 = 4\theta^2$.

$A = \frac{1}{2}\int_0^{\pi} 4\theta^2\,d\theta$.

$A = \frac{4}{2} \times \left[\frac{\theta^3}{3}\right]_0^{\pi}$.

$A = \frac{4}{6} \times (\pi)^3$.

Therefore $A = \frac{2}{3}\pi^3$.

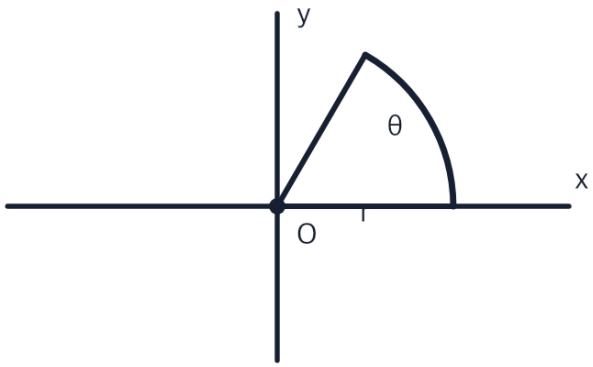
Teacher answers & working

Polar Coordinates and Polar Curves · Area in Polar Coordinates

ALG-294F-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1

Find the area of each polar sector.



Find the area of the sector bounded by r = 6, θ = 0, and θ = π/6.

Answer:

3π

Working:

Use $A = \frac{1}{2}\int r^2\, d\theta$.

Here r = 6, so $r^2 = 36$.

$A = \frac{1}{2}\int_0^{\pi/6} 36\, d\theta$.

$A = \frac{1}{2} \times 36 \times \pi/6$.

Therefore $A = 3\pi$.